

Project Documentation in Data Visualization & Storytelling (S26)

Group A

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Topic:

Affordability of leisure activities for students in Hamburg

Can students like Emma still afford the same leisure opportunities today as Sarah did five years ago?

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What are the main messages of the story, and how do the visualizations in your project contribute to these main messages?

Main message of the story

- Student life in Hamburg has become harder to afford, not just because prices have generally increased, but because rent, housing, daily expenses, and leisure activities have all become more expensive at the same time.
- The main question of our story is therefore not only *whether* everything has become more expensive (that is obvious), but *how much money is actually left for leisure* after all necessary costs are covered.
- We wanted to show what students can still afford with their remaining budget and which leisure activities are still realistic options.
- The story should not only highlight financial limitations, but also show possible opportunities for students to still enjoy free time in Hamburg.

How the visualizations contribute

- The **first visualization** shows the starting budget/income, so readers can see how much money students initially have available.
- The **second visualization** shows the first major expenses, especially living costs such as rent.
- The **third visualization** shows the remaining life expenses, so it becomes clear how much of the original budget has already been used up.
- The **next visualization** shows what is actually left for leisure after all essential expenses.
- The **line chart** on leisure prices shows how the prices of different leisure activities have developed over time.
- The **final chart** creates the “aha moment” by showing how much of each leisure activity students can still enjoy with their remaining budget.
- Together, the visualizations build a clear narrative from income → expenses → remaining money → leisure affordability.

Ending of the website

- We added a section with free or low-cost opportunities in Hamburg at the end of the website.
- This section works as a small guide for students.
- It shows that even with limited money, there are still ways to enjoy the city.
- It gives the story a more practical and positive ending.

Design Decisions and Reflection and possible improvements

Budget Comparison: Waterfall-Chart (Sophie Harre)



Goal of the Visualization and Choice of Chart Type

- The Chart is responsible for the budget comparison, which is the central aha moment of our data story
- Main Goal: show how the monthly budgets of Sarah (2016) and Emma (2021) are structured
- Focus is not only on single numbers, but on the relationship between income, expenses, and remaining budget
- Visualization shows why students can have less money available despite a higher income
- I chose a waterfall chart because it clearly shows changes within a total value
- It visualizes step by step how income and expenses affect the final budget
- Makes complex financial relationships easier to understand
- Chart supports our data story by showing the process from income to remaining budget
- Highlights our main message: a higher income does not automatically mean more financial freedom

Structure, Layout and Design

- **X-axis:** represents the different budget categories (income, rent, living costs, child benefit, and remaining budget), guiding readers through the monthly budget step by step.

- **Y-axis:** represents the monetary values in euros, making it easy to compare how each category increases or decreases the available budget.
- Both charts use the same structure, scale, and design to allow a direct comparison between Sarah and Emma
- Visualization uses a clear comparison approach to make changes between 2016 and 2021 visible
- Different colours for Sarah and Emma provide visual guidance and help users distinguish between the two budgets
- The colours were adapted to the overall website design and create a consistent visual style throughout the data story
- Short title and introduction provide context and connect the chart to the overall story
- Design was intentionally kept simple so that the focus remains on the data instead of unnecessary visual elements

Alternative Design

- A bar chart could have been used as an alternative because it is effective for comparing individual values -> However, a bar chart would mainly show differences between categories and would not explain the process behind the budget development
- It would be more difficult to understand how income and expenses gradually lead to the final remaining budget
- The waterfall chart is therefore more suitable because it shows the development and structure of the budget
- Supports the storytelling purpose of the visualization more effectively

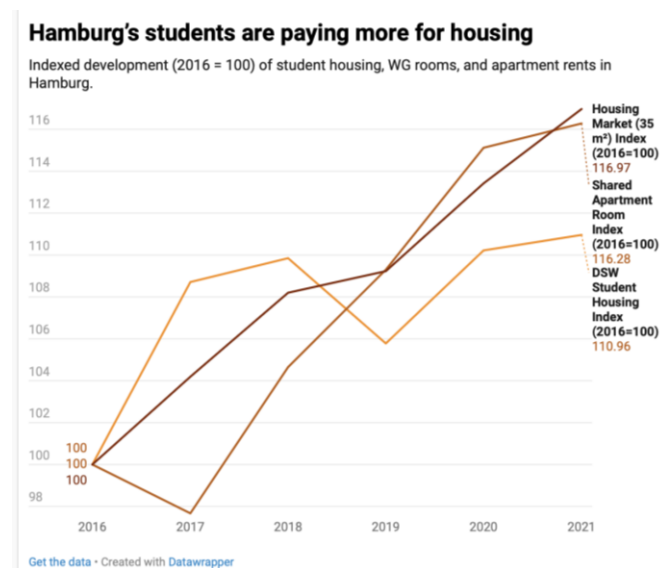
Technical Implementation, Further Development and Improvements

- Flourish provided a good basis for creating the visualization but also had some limitations regarding design options and interactivity
- With stronger programming and web design skills, the visualization could have been developed in a more interactive way
- One possible improvement would be combining both budgets into one interactive chart to make the comparison even easier
- An animation between 2016 and 2021 could highlight the changes over time more clearly
- Users could, for example, adjust individual costs and directly see how the remaining budget changes
- This would allow users to actively explore the data instead of only receiving information from the visualization

- Different income sources, such as BAföG, part-time jobs, or working student jobs could also be included

-> This would create a more realistic picture of students' financial situations and improve the overall message of the data story

Indexed Line Chart- Amal Saib



Goal of the Visualization

- Show how student housing costs in Hamburg have changed over time.
- Make it easy to compare three different housing options (student housing, WG rooms, 35 m² apartments).
- Help viewers quickly understand which option became more expensive and how strong the increase was.

Structure, Layout and Design

Chart Type

- I chose a line chart because it shows changes over time clearly.
- The chart includes three lines, so viewers can easily compare the three housing options.
- A line chart makes trends easy to see and understand.

Visual Encoding

- X-axis: shows the years → the time steps are even and easy to follow.
- Y-axis: shows index values (2016 = 100) → focuses on how much each housing type changed, not on the actual price.
 - Using an index makes the three categories fairly comparable, even though their real prices are very different.
 - Index numbers at the end of each line help viewers quickly see the difference between the three housing types.

- All lines have the same thickness → avoids giving one category more visual weight.

Color Choice

- We decided to use orange for visualizations that relate to both Emma & Sarah
 - I used a **sequential orange color scale with three shades**.
 - This helps viewers see right away that the chart compares three related housing options.
 - The monochrome orange-brown palette looks warm, calm, and easy to read.
 - The darkest shade shows the strongest increase (35 m² apartment market).
 - The medium shade shows the middle increase (WG room).
 - The lightest shade shows the smallest increase (DSW student housing).
- No bright or mixed colors → keeps the chart simple and supports the message: “Costs are rising.”

Overall

- The combination of line chart, indexed values, and simple color scale makes the visualization clear, fair, and easy to understand.
- Viewers can quickly see how each housing type changed over time.

Alternative Design Approach

Absolute values instead of index

- The three housing types have very different starting prices.
- The most expensive one would dominate the chart.
- It would be hard to compare the actual *change* over time.

Bar chart instead of line chart

- Bars make time trends look “blocky” and disconnected.
- It becomes harder to compare three categories across several years.
- The smooth development over time would be lost.
- Year-to-year percentage change
- Only shows small yearly changes.
- The bigger story (“strong increase since 2016”) would not be visible.

Area chart

- The colored areas would overlap.

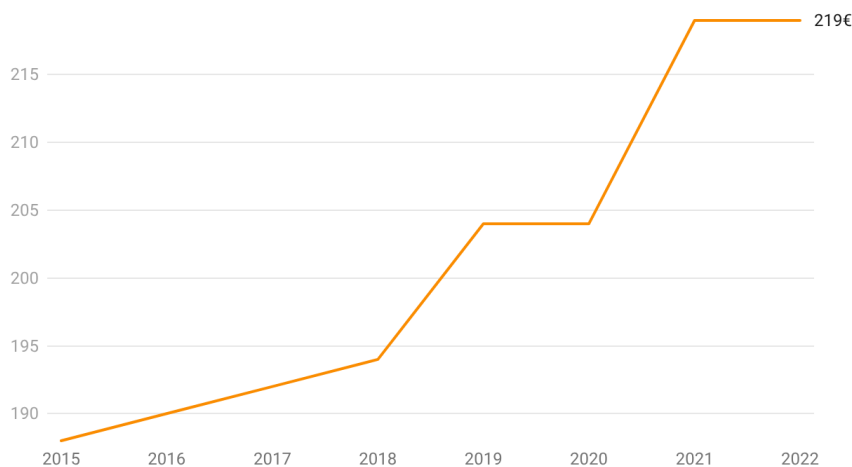
- Viewers might not understand which area belongs to which housing type.
- Comparison becomes confusing.

Dual-axis chart

- Two Y-axes make the chart harder to read.
- Different scales can mislead viewers.
- Often seen as unprofessional or confusing.

Child Benefit Chart – Noelia Boßmann Otero

child benefit & child tax allowance amounts (2015–2022)



Grafik: Noelia Boßmann Otero • Quelle: Sozialpolitik-aktuell.de • Erstellt mit Datawrapper

Goal of the Visualization and Choice of Chart Type

- Responsible for explaining how child benefit developed over time and how it contributes to students' monthly income.
- The goal of the visualization was to illustrate the increase in child benefit between 2015 and 2022, providing context for the financial comparison between Sarah and Emma.
- Although Sarah and Emma only represent the years 2016 and 2021, the complete timeline allows users to understand that the increase was part of a gradual political development rather than a one-time change.
- We chose a line chart because the primary objective was to visualize the development of child benefit over several years.
- A line chart is particularly suitable for displaying trends over time, making it easy to identify periods of gradual increases as well as years in which the amount remained unchanged.
- The visualization prepares users for the following budget calculations by explaining why Emma receives a higher amount of child benefit than Sarah.
- At the same time, it supports the central message of the data story: although financial support increased over the years, this increase alone was not sufficient to offset rising living and housing costs.

Structure, Layout and Design

- The x-axis displays the years 2015 to 2022, while the y-axis shows the monthly child benefit amount in euros.
- Showing the complete time period instead of only the two relevant years provides additional context and allows users to understand the overall development of child benefit over time.
- The line was coloured orange because the visualization refers equally to both Sarah and Emma rather than to one individual person.
- Orange was therefore chosen as a neutral colour that represents the topic itself instead of either sister.
- The chart uses a simple layout with a single line, allowing users to follow the development over time without visual distractions.
- The title clearly communicates both the topic and the observed time period, helping users immediately understand what the visualization represents.
- Positioned between the income section and the expense section, the visualization provides the necessary basis before the following budget calculations.

Alternative Design

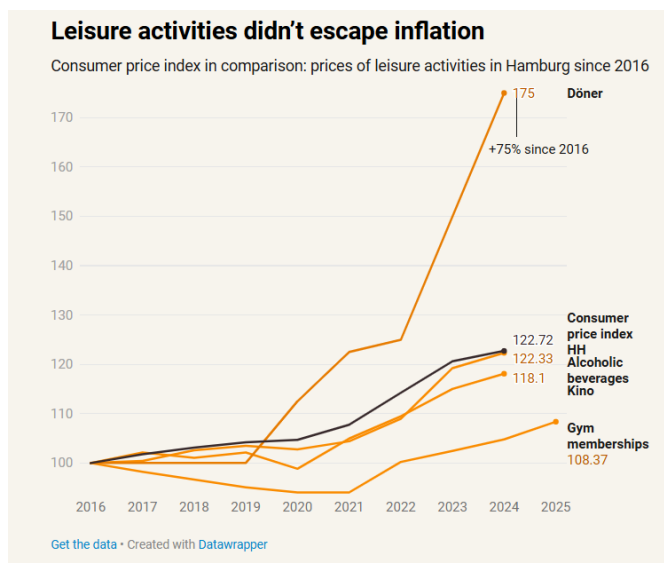
- An alternative would have been a bar chart, where each year is represented by an individual bar.
- A bar chart would make the exact differences between individual years easier to compare, especially because child benefit only changed in selected years.
- However, since the main purpose of the visualization is to show the overall development over time, a line chart was considered the more appropriate choice.

Technical Implementations, Further Development and Improvements

- The visualization was created in Flourish, which provided a straightforward way to display the development over time while keeping the design consistent with the rest of the website.
- Because of the limited customization options, the chart mainly focuses on presenting the trend without additional interactive features.
- A future improvement would be interactive tooltips that explain why child benefit increased in specific years and which policy decisions led to these changes.

- Another improvement would be highlighting 2016 and 2021 directly in the chart, as these are the years relevant to Sarah's and Emma's budgets. This would strengthen the connection between the timeline and the following budget comparison.
- It would also be possible to add a small annotation showing the exact increase between the sisters' starting years (190 € to 219 €) so that users can immediately relate the historical development to the later calculations.
- Finally, the chart could be linked interactively with the waterfall charts so that users can directly see how changes in child benefit affect the sisters' overall monthly budgets, creating a stronger connection between the individual visualizations within the data story.

Price Development of leisure activities - line chart by Lola De Silva



Goal of the Visualization and Choice of Chart Type

- I created the line chart showing the price developments of different leisure activities in Hamburg.
- Our group first decided together which chart type would be most suitable for each visualization, while the detailed design choices were made individually.
- I chose a line chart because the main purpose was to show how prices changed over time and to make the different leisure activities comparable.
- Line charts are especially effective for showing continuous change, long-term trends, and differences in development over time.

Structure, Layout and Design

- The x-axis represents the years, while the y-axis shows the index value.
- This made it possible to see not only that prices changed, but also how strongly they developed relative to the base year 2016.
- Using an index was useful because the original euro values differed greatly and were difficult to compare directly.
- By converting the data into an index with the help of AI, I brought the different price series onto a common scale and focused on relative rather than absolute change.
- From a visual encoding perspective, the chart is metric and ratio-based rather than nominal or ordinal.
- The index uses 2016 as a reference point with the value 100, so values above 100 indicate increases and values below 100 indicate decreases.

- The chart uses multiple visual encodings: position on the axes shows the time trend, while color distinguishes the categories
- I used multiple lines for the different price categories, including cinema, gym, döner, alcoholic beverages, and the Hamburg Consumer Price Index.
- To avoid visual overload, I kept the design simple and used orange for the leisure activities and black for the Consumer Price Index.
- This created a clear visual hierarchy, with the black line standing out as the main reference point.
- I also used direct labels at the end of the lines instead of a separate legend, which makes the chart easier to read, especially on a website.
- Since our narrative follows Sarah and Emma, the visualization needed to be accessible and not overly technical.
- It supports the story by making the price developments easy to understand for a general audience.

Alternative Design Approach

- An alternative design approach would have been to use small multiple line charts or separate charts for each category.
- That could have improved readability for individual trends, especially because the döner line rises much more sharply than the others.
- However, we decided against this because our main goal was comparison.
- A single multi-line chart was better suited to showing that prices changed at different speeds and therefore affect affordability differently.
- Another possible improvement would have been to conduct a survey asking students about their leisure activities and then selecting the three most common ones.
- That would have made the chart more audience-oriented, although it was not necessary for the overall message of the project.

Split bar chart(s) – Leon Hämling



Goal of the Visualization and Choice of Chart Type

- My goal was to show the distribution of monthly costs of living for Emma/ Sarah in 2016/2021
- And highlight which cost categories contribute most to total living costs
- I chose a split bar chart because it allows individual categories and the overall total to be in an easy-to-read format
- It enables direct comparisons between different expense categories and to one another (between 2016 and 2021)

Structure, Layout and Design

- Categories are arranged from left to right - each category is represented by a separate bar and is labelled directly
- The total costs of living (added together) is separated visually with a black divider
- A pink colour scheme was used to represent Sarah – a green colour scheme was used to represent Emma --> maintain visual consistency throughout the website
- I tried to use a “minimalistic” design, so the reader isn't overloaded with information

Alternative Design Approach (and why I did not choose it)

- First, I considered using a pie chart:
 - But rejected it because pie charts can make precise comparisons between categories more difficult and mostly a pie chart is used to show percentages/ an amount of something
- Then I considered using a stacked bar chart:
 - But rejected it as well because again individual expense categories would have been harder to compare visually
- The now chosen design allows every category to remain clearly visible while still showing the overall total --> you can see both, the comparison of the development of each categorie in 2016/2021 and the comparison of the total

Technical Implementations

- The chart was created using Datawrapper
- I imported monthly cost data manually: data comes from the Studierendenwerk from a “Studentenbefragung von DSW Deutschland”:
https://www.studierendenwerke.de/fileadmin/api/files/Soz22_Hauptbericht.pdf
- I applied custom colours, matching the Sarah/Emma colour palette used throughout the website
- And used category labels directly above the bars to avoid the need for a legend

Further Development and Improvements

- To explain each category more, I could add tooltips with further information about each category (explanation of what mobility costs include)
- Include percentage shares of the overall budget
- Add interactive comparison options between multiple years

What were the biggest challenges in designing the visualizations, and how did you overcome them?

1. Challenge: Finding and selecting the **right data** to support our central research question
 - We had to decide which information was truly relevant to show the impact of increasing living costs on students
 - The goal was not just to collect a large amount of data, but to choose data that clearly supports the main message of our data story
2. Challenge: Transforming the collected data into a clear and understandable visual representation
 - We had to consider which chart types were most suitable for each statement and how users could quickly understand the key message
 - automatically lead to more remaining money
3. Challenge: Make sure that the two characters, Sarah and Emma, were visually easy to distinguish
 - This was achieved through the use of different colours and images, creating a clear connection between the person and the data
 - Visualizations directly related to Sarah and Emma use their assigned colours, while charts that are independent from the two characters were designed more neutrally
4. Challenge: Creating a consistent overall design despite using different chart types
 - Through shared design decisions, such as colours, fonts, and layout structures, all visualizations appear as parts of one connected story

Looking back, the main challenge was not only creating individual charts, but combining data, visualization, and storytelling in a way that makes our central question understandable and supports the main message of our data story.

How would you further develop your project if you had more time/ resources?

Data:

- More recent data
- Getting some missing data / more hamburg related data --> working with the data for the budget prices, we found out that its difficult to find some data specifcly for hamburg instead of nationwide
- Own big data collection, collect own data through surveys on the website

Visual design:

- moving graphics (Sarah and Emma)
- More flexible structure --> more variety regarding design (eventually asymmetrical structures)
- Charts in the same design as the website, Sarah, and Emma --> more on theme, more fitting in the sotryline/reading process --> more homogeneous
- Adding quotes of reals students about their struggels and opinions about their budget and experiences in Hamburgs student life (eventually from both time spans --> deeper comparison between then and now)

Adding more aspects:

- Interactive calculator, with which readers can get their own package of leisure activities by typing in their amount of monthly budget
 - An additional function could be a kind of month planner that helps planning your own budget and deducts the prices of the leisure activities you would like to do
- Add a section where you can compare all data from Sarah and Emma with their “friends” who are living in other cities -
 - Example: Tom tells you about his life in Berlin, studying the same bachelor's degree but having to deal with even less money because of higher rents
 - Example: Patricia studies in Leipzig and has a higher budget than Sarah, Emma and Tom, so she talks about different and more leisure activities she can spend her money for

Use of AI

Use of Chat GPT

- Creating graphics/ photos of our main characters Sarah and Emma
- Photos created: 6 different graphics

Use of Perplexity

- For the line charts we used AI to calculate the index of the absolute values, because the scales were different. Through the index we could compare the relative values with each other

AI used for creating the website

- Help with creating the Website
 1. Creation of a template (based on our storyline) without AI on Canva
 2. Use of Claude: building the main structure for our Website as html and css files
 3. Implementing our template with long AI prompts with Microsoft CoPilot
 4. Adjusting last details and nuances with Microsoft CoPilot and without AI

Sources

- Income:
Studierendenbefragung DSW Deutschland:
https://www.studierendenwerke.de/fileadmin/api/files/Soz22_Hauptbericht.pdf
- Rent:
Zahlenspiegel Einzelbericht 2016-2024:
https://www.studierendenwerke.de/fileadmin/user_upload/dsw_zahlenspiegel_2016_2017_internet.pdf
- Mietpreise Wohnungsmarkt:
Statista - Data from: Kiel Institut für Weltwirtschaft; Value AG (Marktdatenbank):
<https://de.statista.com/statistik/daten/studie/971531/umfrage/mietpreise-auf-dem-wohnungsmarkt-in-hamburg/>
- WG-Gesucht Mietpreisentwicklung:
Anfrage an: pr@wg-gesucht.de
- Living costs:
Studierendenbefragung DSW Deutschland:
https://www.studierendenwerke.de/fileadmin/api/files/Soz22_Hauptbericht.pdf
- child benefit:
Höhe von Kindergeld und Kinderfreibeträgen 1998 - 2025
https://www.sozialpolitik-aktuell.de/files/sozialpolitik-aktuell/_Politikfelder/Familienpolitik/Datensammlung/PDF-Dateien/tabVII14.pdf
- Inflation increase (Lola):
Statista - Verbraucherpreisindex in Hamburg von 2015 bis 2024 – data from: Statistisches Bundesamt:
<https://de.statista.com/statistik/daten/studie/1128345/umfrage/entwicklung-des-verbraucherpreisindex-hamburg/>
- Mitgliedsbeiträge für Fitnessstudios in Deutschland 2025
Statista - Data from: EuropeActive; Deloitte; DSSV; Deutsche Hochschule für Prävention und Gesundheitsmanagement
<https://de.statista.com/statistik/daten/studie/6974/umfrage/mitgliedsbeitraege-fuer-fitness-studios-im-europaeischen-vergleich/>
- Durchschnittlicher Kino-Eintrittspreis bis 2029:
Statista – data from: PwC
<https://de.statista.com/statistik/daten/studie/165819/umfrage/durchschnittlicher-kino-eintrittspreis-seit-2005/>
- Entwicklung der Verbraucherpreise von alkoholischen Getränken in Deutschland bis 2025:
Statista – data from: Statistisches Bundesamt
<https://de.statista.com/statistik/daten/studie/305680/umfrage/entwicklung-der-verbraucherpreise-von-alkoholischen-getraenken-in-deutschland/>

- Preisentwicklung Döner in Hamburg 2016 bis 2024 in €:
<https://medium.com/@jandegener/entwicklung-der-dönerpreise-in-deutschland-2016-2024-de-b7f9bb155c7d>